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Numerical simulation of fully-developed compressible flows over wavy surfaces

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ABSTRACT

Rough surfaces are common on high-speed vehicles, for example on heat shields, but compressibility is not usually taken into account in the flow modelling other than through the mean density. In the present study, supersonic fully-developed turbulent rough wall channel flows are simulated using direct numerical simulation to investigate whether strong compressibility effects significantly alter the mean flow and turbulence properties across the channel. The simulations were run for three different Mach numbers $M = 0.3$, 1.5 and 3.0 over a range of wall amplitude-to-wavelength ratios from 0.01 to 0.08, corresponding to transitionally and fully rough cases respectively. The velocity deficit values are found to decrease with increasing Mach number. It is also found that at Mach 3.0 significant differences occur in the mean flow and turbulence statistics throughout the channel and not just in a roughness sublayer. These differences are found to be due to the presence of strong shock waves created by the peaks of the roughness elements.

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1. Introduction

Turbulent flows over rough surfaces are common in many different applications and have been the subject of numerous investigations. Due to the wide range of possible surface topographies, evaluating and predicting the effect of the surface on a flow has relied upon empirical correlations based on experimental studies. The early work of Goddard (1959) on commercial sand-paper roughness up to Mach 5 was the first to suggest that the effect of compressibility is indirect and that the skin-friction drag is only a function of the roughness Reynolds number, as with incompressible flow cases. This Mach number independence should not be surprising in this case since Goddard observed no wave drag component for the range of rough surfaces tested. The work of Berg (1979), however, identified compressibility effects in a Mach 6 flow over 2-D transverse bar elements. Berg (1979) compared his results against an effective roughness correlation developed by Dirling (1973) for low-speed flows and found a factor of three error between the results, suggesting that there are significant compressibility effects. More recent works of Latin and Bowersox (2000, 2002) have shown shock and expansion waves, formed by roughness elements protruding out of the sublayer for a Mach 2.9 flow. The authors suggest these waves introduce a connection between the inner and outer layers of the boundary layer, which suggests that it may be possible for roughness elements to cause large changes in the turbulent flow characteristics. Further experimental work by Ekoto et al. (2008) sought to deliberately cause strong compressibility effects by using diamond-shaped elements in a Mach 2.9 flow. They found that these strong compressibility effects had a significant effect on the turbulent flow statistics, such as the Reynolds shear stress, across the entire boundary layer.

Previous computational studies of compressible channel flow have been primarily limited to the study of smooth channels. Coleman et al. (1995) used Mach numbers of 1.5 and 3.0 and a Reynolds number of 3000 to show that for these Mach numbers there is comparatively little effect on the turbulent flow between compressible and incompressible flows. Lechner et al. (2001) used the same smooth channel set-up at a Mach number of 1.5 to show that the compressibility may have some effect on the turbulent production and dissipation near the wall in areas of high density.

Studies of flow over 2D wavy walls have been primarily for incompressible flows. Works by De Angelis et al. (1997) and Cherukat et al. (1998) used DNS to study a wavy surface previously used in an experimental study by Hudson et al. (1996). In all three studies, a wavy surface with an amplitude-to-wavelength ratio $\alpha/\lambda = 0.05$ was used, with a bulk Reynolds number of 3380. Results indicated that the mean streamwise velocity profiles across the surface wavelength only varied when near the wall, below one-third of the channel half-height, so that wave-induced flow variations are small once in the outer-layer of the flow. Skin friction profiles show that the flow undergoes separation in the surface trough, although Cherukat et al. point out that these separations

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vary both in time and space, with large variations present across the span. Sun et al. (2011) performed a DNS on this same surface at a Mach number of 1.5. Slices of the domain showing pressure contours indicate the potential for shocks to exist within the channel, since wave structures are visible, however no values for the pressure are presented, therefore the strength of these features is unknown. Also shown are velocity profiles taken at the peak and trough of the wavy surface which show that, although the streamwise velocity is the same as in the incompressible cases, the wall-normal velocity experiences some changes. These features suggest the potential for strong compressibility effects generated by the surface to alter flow properties in the outer-layer of the flow. Other works looking at the compressibility effects of rough surfaces have been focused on transitional flows, such as Redford et al. (2010) and Bernardini et al. (2012), rather than looking at a fully developed turbulent flow over distributed roughness.

The present paper presents a systematic study of turbulent compressible channel flow over a wavy wall. Both the surface wave amplitude, wavelength and the Mach number are varied to study compressibility effects over a range of surfaces, from smooth to fully rough.

2. Simulation set-up

2.1. Governing equations

A Direct Numerical Simulation (DNS) code is used to generate the results presented. This code solves the full 3D, compressible Navier–Stokes equations in the following form

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} = 0, \tag{1}$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_j} = -\frac{\partial p}{\partial x_j} + \frac{\partial \tau_{ij}}{\partial x_j} + F\delta_{i1}, \tag{2}$$

$$\frac{\partial \rho E}{\partial t} + \frac{\partial (\rho E + p) u_j}{\partial x_j} = \frac{\partial u_i \tau_{ij}}{\partial x_j} - \frac{\partial q_j}{\partial x_j} + u_i F \delta_{i1}, \tag{3}$$

where the stress tensor and heat flux are

$$\tau_{ij} = \frac{\mu}{\mathrm{Re}}\left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j} - \frac{2}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij}\right), \tag{4}$$

$$q_j = -\frac{\mu}{(\gamma - 1)M^2 \mathrm{RePr}}\frac{\partial T}{\partial x_j}. \tag{5}$$

All the parameters are non-dimensionalised by the channel half-height, wall temperature and reference values obtained from the bulk quantities. The viscosity is calculated by using Sutherland's law and forcing terms, F and u_iF, equivalent to an applied streamwise pressure gradient and its associated work term, are included in the momentum and energy equations respectively. A non-slip boundary condition is used for the velocity components whilst an isothermal boundary is applied for the temperature, setting this to 1.0 at the wall.

2.2. Numerical method

The DNS code used for the simulations presented is fourth-order accurate in space, using a central difference scheme for the internal points. The wall boundaries are fully resolved by a body conforming grid and a treatment by Carpenter et al. (1999) is used at these boundaries. A third-order Runge–Kutta method is used for the time integration. To improve stability, an entropy splitting approach by Sandham et al. (2002) was used, along with the TVD shock treatment scheme by Yee et al. (1999) to accurately capture shock waves. The code was parallelised using the MPI library.

2.3. Validation

The code was validated for a smooth channel flow case which included density in the forcing terms as in Coleman et al. (1995). For this case a domain size of $12 \times 2 \times 4$ was used with a grid using $100 \times 140 \times 94$ points in the streamwise, wall-normal and spanwise directions respectively. A uniform grid spacing was used in the streamwise and spanwise directions, whilst a hyperbolic tangent function was used for the wall-normal direction, given by

$$\frac{y(\eta)}{\delta} = \frac{\tanh(b\eta)}{\tanh(b)}, \tag{6}$$

where η is uniformly distributed on $[-1,1]$, $b = 1.5$ is a stretching factor and δ is the channel half height. This resulted in grid spacings of $\Delta x^+ = 19.6$ and $\Delta z^+ = 7.0$ while the minimum wall-normal spacing was $\Delta y^+ = 0.7$. The results for this validation are shown in Fig. 1 for some of the mean and Favre-averaged properties. The Favre-averaged root mean square fluctuation is defined as $\phi'' = \sqrt{\overline{(\phi - \tilde{\phi})^2}}$ where $\tilde{\phi} = \overline{\rho\phi}/\bar{\rho}$ and ϕ represents a fluid variable. Fig. 1a shows very close agreement for all the properties, whilst for the velocity fluctuations in Fig. 1b the results show a difference, which is considered to be sufficiently small.

Fig. 2 shows the two point correlations for the smooth validation case in both the streamwise and spanwise directions at a distance of $1 - |y| = 0.04$. These correlations demonstrate the adequacy of the domain size which agree with the correlations in Coleman et al. (1995).

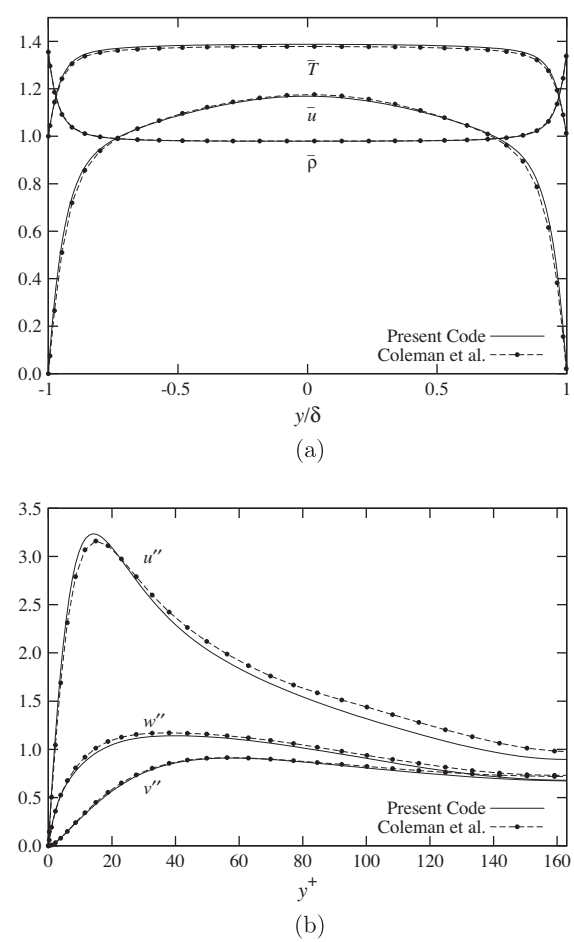


Fig. 1. Results of the validation study (a) mean flow properties $\bar{u}$, $\bar{\rho}$ and $\bar{T}$ and (b) Favre-averaged velocity fluctuations.

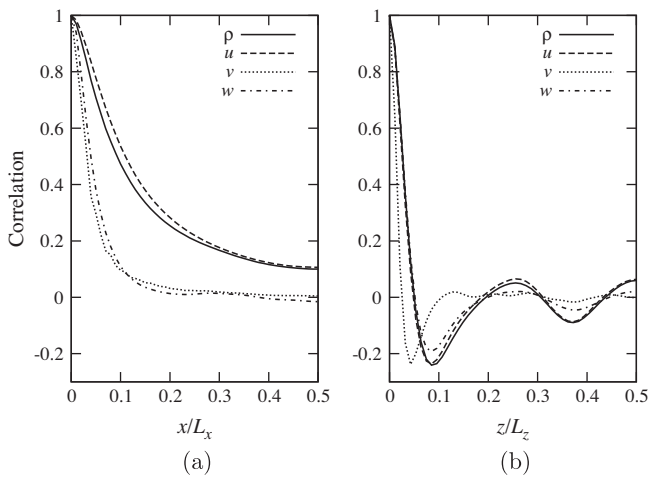


Fig. 2. Two point correlations at $1 - |y| = 0.04$ for the validation case in (a) streamwise and (b) spanwise directions.

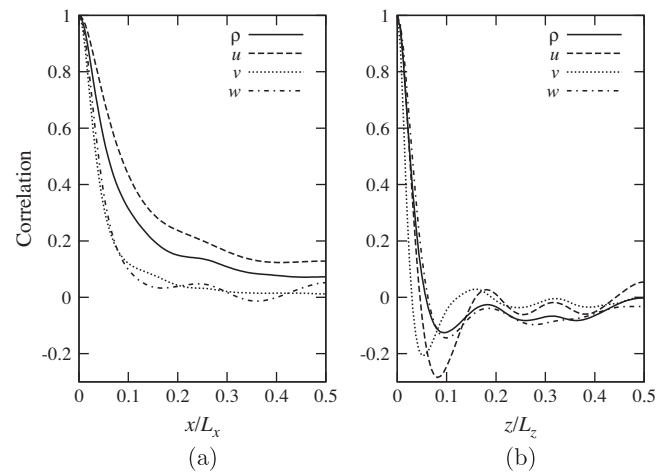


Fig. 4. Two point correlations at $1 - |y_{\text{wall}}| = 0.08$ for the validation case in (a) streamwise and (b) spanwise directions.

2.4. Physical geometry

The wall surfaces investigated were sinusoidal waves in the streamwise directions with no surface variation across the span on both the upper and lower walls. The wall surfaces were generated by

$$y(x) = y_{\text{wall}} + \alpha \cdot \cos\left(2\pi \frac{n_x}{L_x} x\right), \tag{7}$$

where y_{wall} is the mean wall location, n_x is the integer number of waves in the domain, which varied as $n_x = 2$, 4 and 8, L_x is the domain length and the surface amplitude took on three values, $\alpha = 0.00$, 0.04 and 0.08. An example of the surface is shown in Fig. 3. Each surface was tested at Mach numbers of 1.5 and 3.0, with a reference subsonic case of $M = 0.3$ also simulated. The $M = 0.3$ and 1.5 cases were run at a bulk Reynolds number of $\text{Re}_b = 3000$, while the $M = 3.0$ cases were run with $\text{Re}_b = 4880$ to ensure that the centreline Reynolds number was similar for the different Mach number cases. The computational domain size used was $L_x \times L_y \times L_z = 8\delta \times 2\delta \times 4\delta$ where $\delta = 1$ is the dimensionless channel half-height. The simulations used $240 \times 187 \times 160$ grid points in the streamwise, wall-normal and spanwise directions respectively for $\text{Re}_b = 3000$, and $390 \times 303 \times 260$ for $\text{Re}_b = 4880$.

Fig. 4 shows the two-point correlations for a wavy wall case where $\alpha = 0.04$ and $n_x = 2$ for $M = 1.5$. The correlations were taken at a height of $1 - |y_{\text{wall}}| = 0.08$, which is twice the height of the surface wave used here. These correlations show that the domain size used was adequate since the correlations tend towards the same values as in the smooth wall case.

3. Results

The results of the simulations will be divided into two categories. First, the results for a varying surface wave amplitude for a constant surface wave-length will be considered. Then the effect of the surface wavelength will be looked at for the two different wavy wall amplitudes. Each study includes the effect of Mach number. The results in the next sections are all with the forcing terms given in Section 2.1 rather than the validation case forcing terms of Coleman et al. (1995).

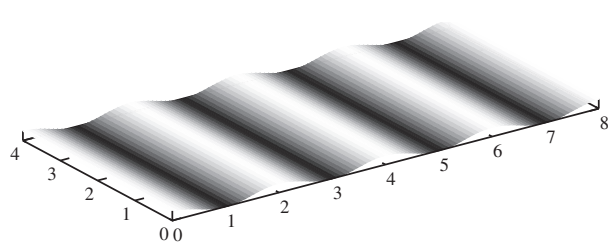


Fig. 3. Example of the wavy surface used grey shaded with surface height.

3.1. Effect of amplitude

Table 1 shows the cases used to investigate the effect of changing surface amplitude, along with the case designations by which each case will be referred to. The first letter of the designation refers to the Mach number of the case, with S representing the subsonic $M = 0.3$ case, L representing the low supersonic case of $M = 1.5$ and H the higher supersonic case of $M = 3.0$. The first number represents the surface wavelength given by $\lambda = L_x/n_x$ while the second number refers to the surface wave amplitude.

Fig. 5 shows the van Driest transformed mean velocity profiles, which is defined as

$$u^+_{\text{VD}} = \int_0^{\bar{u}^+} \left(\frac{\bar{\rho}}{\bar{\rho}_w}\right)^{1/2} \mathrm{d}\bar{u}^+, \tag{8}$$

where ρ_w is the density at the wall and $\bar{u}^+$ is the mean streamwise velocity normalised by the smooth wall friction velocity. The van Driest profiles have been averaged in both the streamwise and spanwise directions as well as over time. The origin for the wall-normal co-ordinate is taken to be the mean wall location, which is clearly defined for this sinusoidal surface at $y = -1$, coinciding with the wall location for the smooth channel case. The visible kinks in the profiles show the location of the peak of the surface wave. One of the effects of the increased Reynolds number for the $M = 3.0$ case is visible in the location of these peaks, since when non-dimensionalised by the viscous lengthscale, the height of the surface for the $M = 3.0$ case is 15% larger, as are the channel

Table 1
Summary table for 2D Wavy wall fixed wavelength cases for $n_x = 4$.

		Mach number		
		0.3	1.5	3.0
α	0.00	S-2-00	L-2-00	H-2-00
	0.04	S-2-04	L-2-04	H-2-04
	0.08	S-2-08	L-2-08	H-2-08

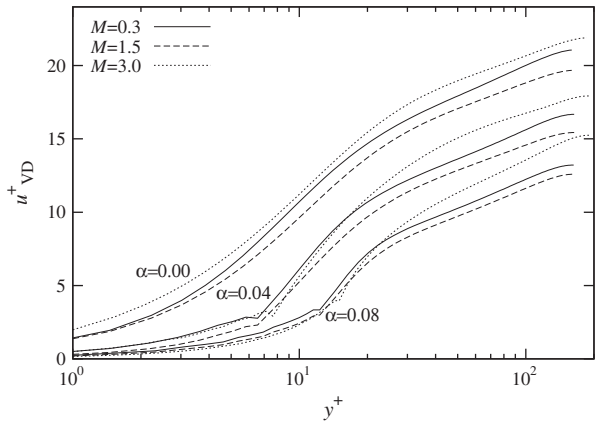


Fig. 5. Van Driest transformed velocity profiles for $M = 0.3$, $M = 1.5$, and $M = 3.0$.

dimensions. The plots show a consistent decrease in the peak velocity value as the surface amplitude is increased for each Mach number corresponding to increasingly rough conditions. The velocity deficit, Δu^+_{VD}, is measured at the centreline relative to the smooth wall case. For the Mach 0.3 case $\Delta u^+_{VD} = 4.38$ and 7.84 for the 0.04 and 0.08 surfaces respectively, for the Mach 1.5 case these are 4.24 and 7.09 while for the Mach 3.0 case these are 4.02 and 6.70. Based on these values we can classify the $\alpha = 0.04$ case as transitionally rough and the $\alpha = 0.08$ case as fully rough. These values are plotted against Mach number for the two surface amplitudes in Fig. 6. From this plot it can be seen that the velocity deficit decreases with increasing Mach number. A difference is observed with respect to the work of Berg (1979) since the van Driest transformed velocity profiles across three different Mach numbers have not collapsed to give the same result. As well as possible Reynolds number effects, it appears that the increased density near the wall for channel flow, associated with the increased Mach number, may be reducing the impact of the roughness upon the rest of the flow.

Owing to the streamwise surface variation, streamwise averaged results can conceal interesting areas in the flow. Therefore Fig. 7 shows the spanwise and time averaged streamwise velocity profiles for each of the cases at the peak and trough locations along the surface. There is a significant difference in the velocity profiles when $\alpha = 0.04$. For the $M = 0.3$ and $M = 1.5$ cases the flow over the trough is always slower than the flow over the peak, although in both cases the profiles follow a similar trend and remain close in value across the whole channel width. In contrast, in the $M = 3.0$ case the flow over the trough is always faster than over the peak.

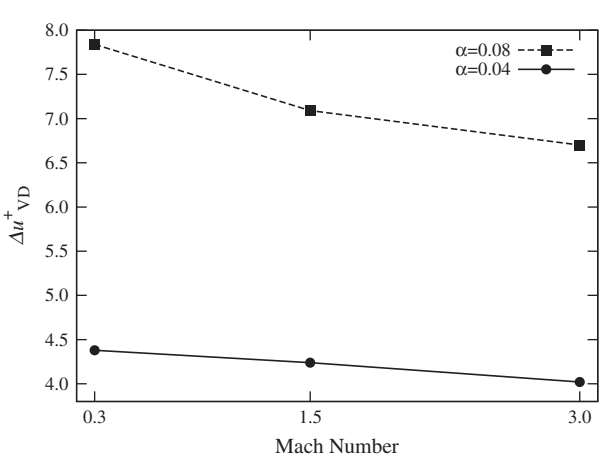


Fig. 6. Comparison of Δu^+_{VD} for different surface amplitudes.

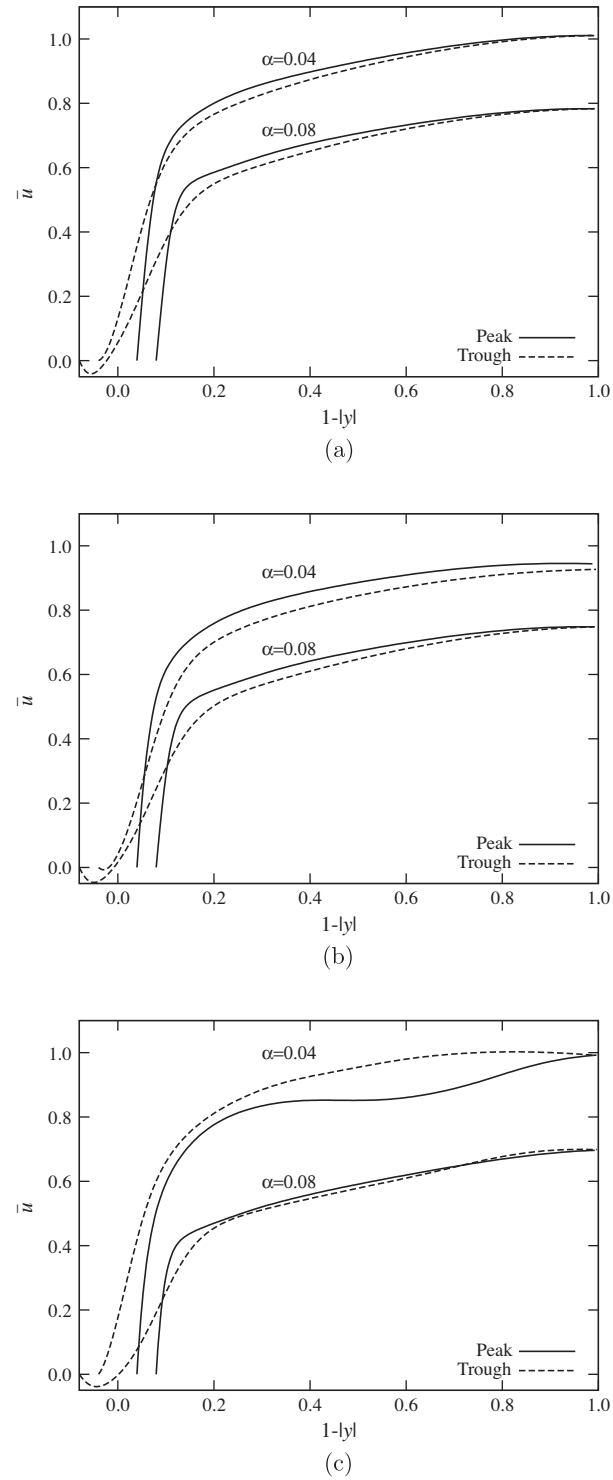


Fig. 7. Mean velocity, $\bar{u}$, profile slices at the peak and trough of the wall for (a) $M = 0.3$, (b) $M = 1.5$, and (c) $M = 3.0$.

The peak profile also has a dip in velocity at $1 - |y| \approx 0.6$, indicating that some significant changes occur in this channel despite the van Driest transformed profiles suggesting a more benign flow field. The $\alpha = 0.08$ surfaces at $M = 3.0$ are more similar, however the velocity profile of the fluid over the trough for the $M = 3.0$ case is steeper near $1 - |y| \approx 0.5$ and flatter near the channel centreline, again suggesting that there is something in the $M = 3.0$ channel not experienced with the lower Mach numbers.

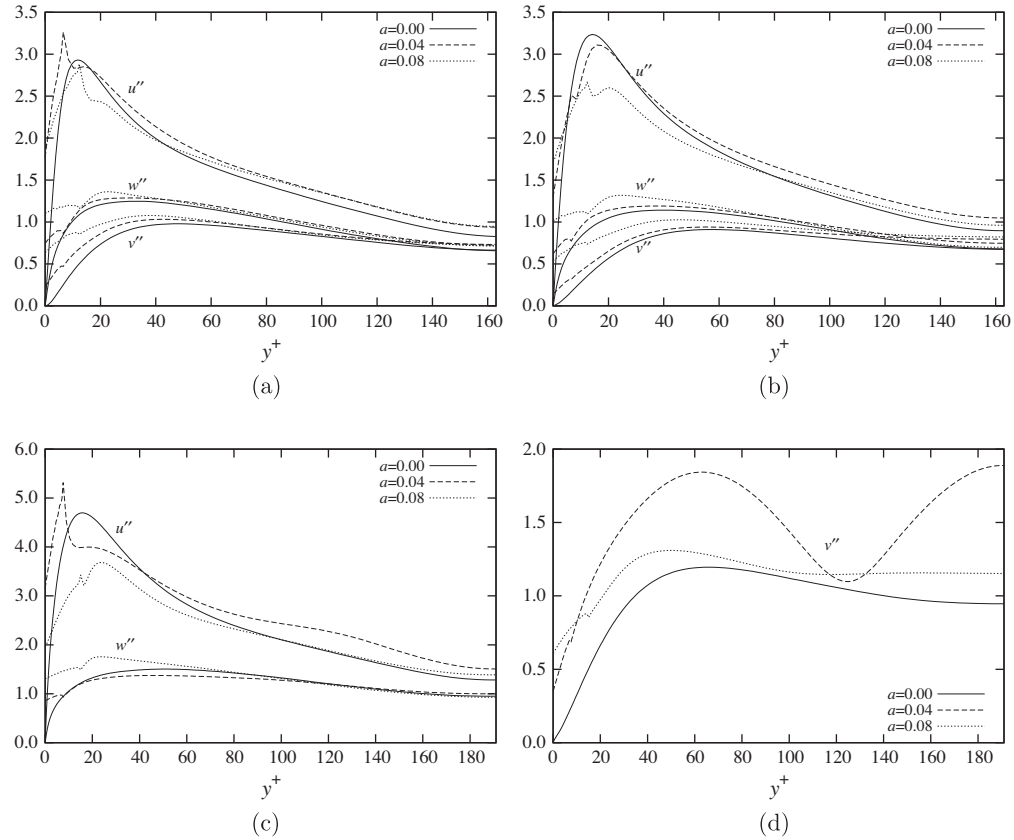


Fig. 8. Mean Favre-averaged velocity fluctuation profiles in wall units for (a) $M = 0.3$, (b) $M = 1.5$, (c) streamwise and spanwise fluctuations for $M = 3.0$, and (d) wall-normal fluctuations for $M = 3.0$.

Fig. 8 shows the time, streamwise and spanwise Favre-averaged velocity fluctuation profiles shown here in wall-units. Owing to the different Reynolds number in the $M = 3.0$ case the y^+ range and fluctuation variable values are greater than found at the lower Reynolds number cases. The streamwise fluctuations, u'', are similar for the $M = 0.3$ and 1.5 cases, shown in Fig. 8a and b respectively. At both Mach numbers the $\alpha = 0.04$ surface tends to have higher values in the channel core, with the main differences near the peak around $y^+ = 20$. The $M = 3.0$ cases in Fig. 8c roughly follow this trend, although owing to the higher bulk Reynolds number the fluctuation values are greater for the $M = 3.0$ cases compared to other Mach numbers. Case H-2-04 experiences a large near-wall peak relative to the smooth wall, whilst case H-2-08 has a reduced peak. Additionally a small bump in the H-2-04 profile can be seen in Fig. 8c around $y^+ = 125$, suggesting that the roughness is having some effect towards the channel centreline for this case.

The spanwise velocity fluctuations, w'', for $M = 0.3$ and 1.5 in Fig. 8a and b show only minor differences, with both wavy wall cases having higher fluctuation levels compared to the smooth wall core. For the near wall peak, the $\alpha = 0.08$ surface has a higher value, but further towards the channel centreline this decreases below the profile for $\alpha = 0.04$. The $M = 3.0$ cases in Fig. 8c show different trends. The $\alpha = 0.04$ surface experiences reduced fluctuations compared to the smooth wall case, whilst the $\alpha = 0.08$ surface follows the same pattern as with the other cases. Towards the channel centreline the two wavy wall profiles once again cross, but in general the spanwise fluctuations only show small Mach number effects.

The wall-normal profiles, v'', show a much more dramatic difference between the different Mach number flows. Again the profiles for the $M = 0.3$ and 1.5 cases in Fig. 8a and b, show the same trends; as the surface amplitude is increased so too are the fluctuation amplitudes. In the $M = 3.0$ case, shown in Fig. 8d, the $\alpha = 0.04$ case is substantially different to any of the other profiles. Two large peaks are clearly visible, one near $y^+ = 60$ and the other the channel centreline, while the profile for the $\alpha = 0.08$ surface appears to remain similar to that of the lower Mach numbers. In order to assess what caused these peaks contour plots of the time and spanwise averaged v'' profiles are plotted in Fig. 9. These plots show that, in contrast to Fig. 8, there are some differences between the $M = 0.3$ and 1.5 cases, with the $M = 1.5$ having more localised peaks located over the surface trough. Fig. 9c clearly shows why the $\alpha = 0.04$ surface has large peaks away from the wall, since the peaks align with strong wave-like features in the flow emanating from just behind the surface peak. The source of these will be covered later in this section.

Since the features found in the wall-normal velocity fluctuation slices have the appearance of shock waves, the divergence of velocity, $\nabla \cdot u$, was calculated for an instantaneous slice of each case. These slices are shown in Figs. 10 and 11. Shock waves are clearly seen in both $M = 3.0$ cases, indicated by the strong compressions across a small area. Comparing cases H-2-04 and H-2-08, the shock strengths are greatest for the $\alpha = 0.08$ surface, although the shock pattern is less regular. It is supposed that since the mean wall-normal velocity fluctuations only showed a significant variation for the H-2-04 case, the shock waves here must be regularly occurring in approximately the same location in order to cause these differences to the statistics. In contrast, since there is almost no change in the H-2-08 wall-normal fluctuations it seems likely that the shock formation is more transient. This is confirmed by Fig. 12 which shows the $\nabla \cdot \bar{u}$ for the $M = 3.0$ cases, where it is clear from the contour scale that the mean compression strength for the $\alpha = 0.04$ case is almost double the $\alpha = 0.08$ case.

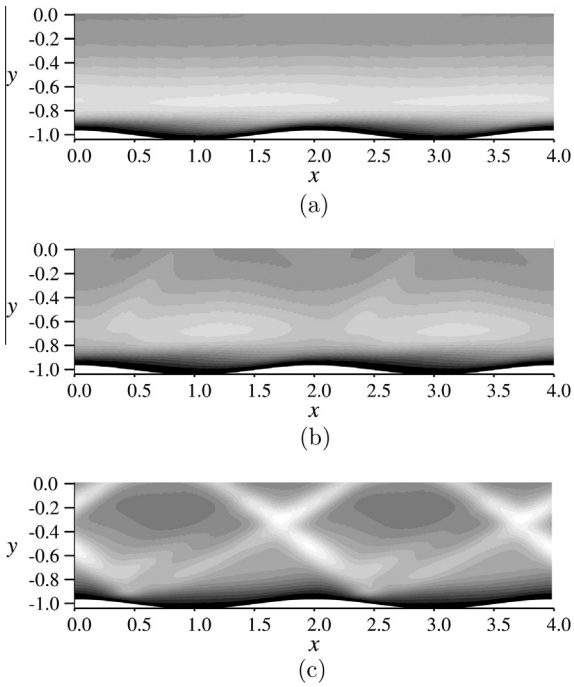


Fig. 9. Favre-averaged wall-normal velocity fluctuation slices in the *xy*-plane for (a) $M = 0.3$, (b) $M = 1.5$, and (c) $M = 3.0$.

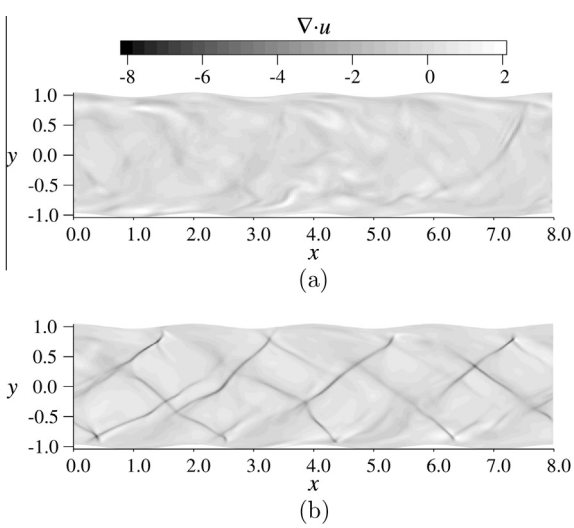


Fig. 10. Instantaneous $\nabla \cdot u$ contours with $\alpha = 0.04$ for (a) $M = 1.5$ and (b) $M = 3.0$.

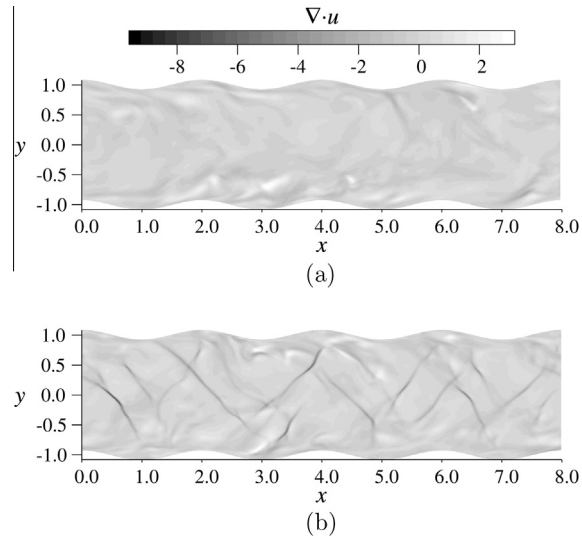


Fig. 11. Instantaneous $\nabla \cdot u$ contours with $\alpha = 0.08$ for (a) $M = 1.5$ and (b) $M = 3.0$.

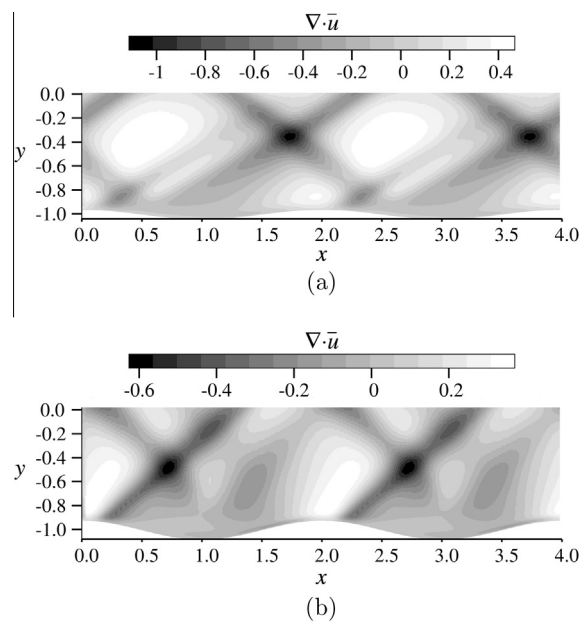


Fig. 12. Contours of $\nabla \cdot \bar{u}$ for $M = 3.0$ for (a) $\alpha = 0.04$ and (b) $\alpha = 0.08$.

Since there are shock waves present in the $M = 3.0$ channels it is useful to observe how these features affect the drag forces upon the surface. This force can be broken down into two components: the net pressure acting on the surface peaks, and the skin friction acting along the surface. The net pressure is calculated from the horizontal pressure force on the upstream side of the surface peak minus the pressure force on the downstream side of the peak. Fig. 13 shows these pressure distributions as a function of surface peak height. It is important to note that, owing to the higher Reynolds number used for the $M = 3.0$ case, the forcing term used to maintain a constant flow rate in the channel was lower. Therefore the absolute values for the $M = 3.0$ cases are reduced in comparison to the other cases. Both plots in Fig. 13 show that the location of the maximum pressure force moves with a change in Mach number. This is most clearly seen in Fig. 13b for $\alpha = 0.08$, where the peak consistently moves towards the surface peak with increased Mach number. This is expected since the upper parts of the surface peak are exposed to higher fluid velocities. Fig. 13a for $\alpha = 0.04$ does not show this trend. From $M = 0.3$ to 1.5 the pressure maximum moves towards the surface peak. However, for the $M = 3.0$ case the pressure maximum has moved downwards again.

The difference in the net pressure behaviour may due to the change in recirculation behaviour in the surface trough. Fig. 14 shows the mean skin friction profile along a single wavelength. Fig. 14a for the $\alpha = 0.04$ surface shows that only the $M = 1.5$ case undergoes separation, whereas the other two cases remain attached throughout. So, in the case of a recirculation region, the interaction between faster moving fluid and the wall is moved further away from the surface trough by the recirculation. Fig. 14b shows the same set of profiles for the $\alpha = 0.08$ surface. These

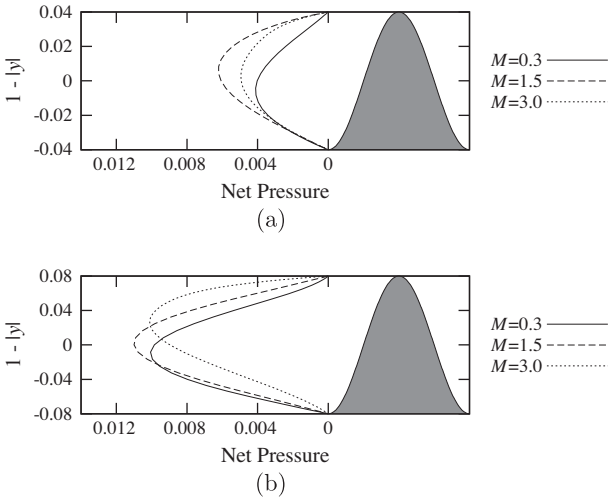


Fig. 13. Net pressure profiles for (a) $\alpha = 0.04$ and (b) $\alpha = 0.08$.

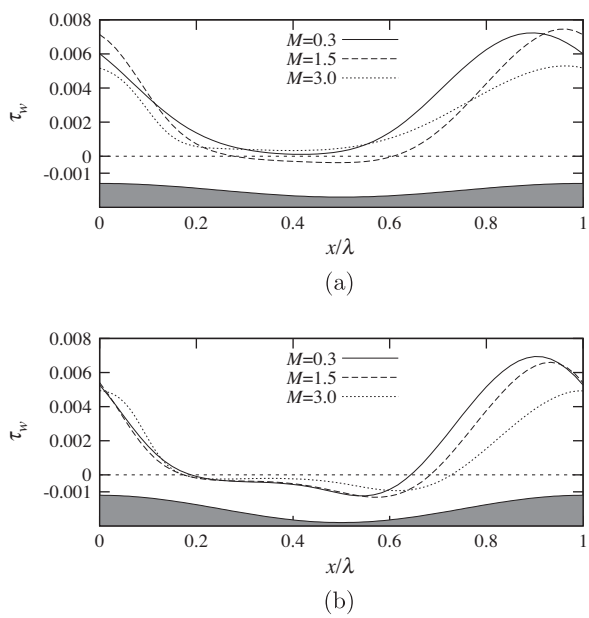


Fig. 14. Skin friction profiles for (a) $\alpha = 0.04$ and (b) $\alpha = 0.08$.

profiles are much more similar, with all surfaces showing flow separation at approximately the same point, although the flows reattach at different points. The reattachment point is delayed with increased Mach number, leading to larger recirculation zones.

Fig. 15 shows the percentage of the total drag force on the surface that can be attributed to the pressure and skin friction forces. For the $\alpha = 0.08$ surface there is a clear trend that with an increase in Mach number the pressure contribution increases. This same trend is not seen for the $\alpha = 0.04$ surface, where instead the pressure contribution decreases for the $M = 3.0$ case. It is believed that this is associated with the lack of a recirculation region previously mentioned in the H-2-04 case, which not only increases the skin friction experienced along the channel, but modifies the pressure distribution. Comparing the two different surface amplitudes it is clear that the $\alpha = 0.08$ surface is a pressure dominated case, whereas the $\alpha = 0.04$ surface is a skin friction dominated case.

In summary, the comparison of two different surface amplitudes for a fixed surface wavelength across the three Mach numbers has shown a mixture of effects. In the $M = 3.0$ cases shock

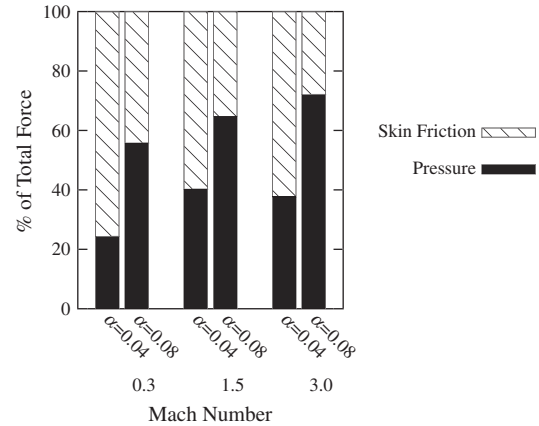


Fig. 15. Percentage of total drag force attributed to pressure and skin friction.

waves were found to exist. The lower surface caused weaker, but more regular shock waves. These regular shock waves significantly alter the turbulent statistics for the H-2-04 case. For example, the peaks of the wall-normal velocity fluctuations are located at the shock interaction points, rather than being localised over the surface. However, these changes appear to have little affect on the mean van Driest transformed velocity profiles, since the surface has a reduced velocity deficit than the equivalent cases at the lower Mach numbers. Therefore it is interesting to investigate whether these phenomena are limited to just this particular surface amplitude and wavelength combination, or if these are generic features.

3.2. Effect of wavelength

Following on from the results presented in the previous subsection, additional surfaces were simulated to further investigate the effect of the wavy surface on the flow. Whereas previously the surface had a fixed wavelength, this value was then varied. A summary of the different cases and the designation given to each case can be seen in Table 2. The format of the designation is the same as with the previous study.

Fig. 16 shows the time, streamwise and spanwise averaged van Driest velocity profiles for each case. The plots show a greater variation in behaviour between the different surfaces wavelengths compared to the change in amplitude study discussed previously. In all except one case, decreasing the surface wavelength increases the velocity deficit. There is one exception to this seen in plot (a) for case S-1-08 which has the least velocity deficit for an $\alpha = 0.08$ surface at $M = 0.3$. This can be explained by considering which roughness regime the surfaces are in. In the work by Jimenez (2004) there is a discussion of k-type and d-type roughness which are differentiated by whether the roughness elements provide a

Table 2
Case designation for wavelength variation cases.

	Mach number	
	1.5	3.0
(a) $\alpha = 0.04$		
λ		
1	L-1-04	H-1-04
2	L-2-04	H-2-04
4	L-4-04	H-4-04
(b) $\alpha = 0.08$		
1	L-1-08	H-1-08
2	L-2-08	H-2-08
4	L-4-08	H-4-08

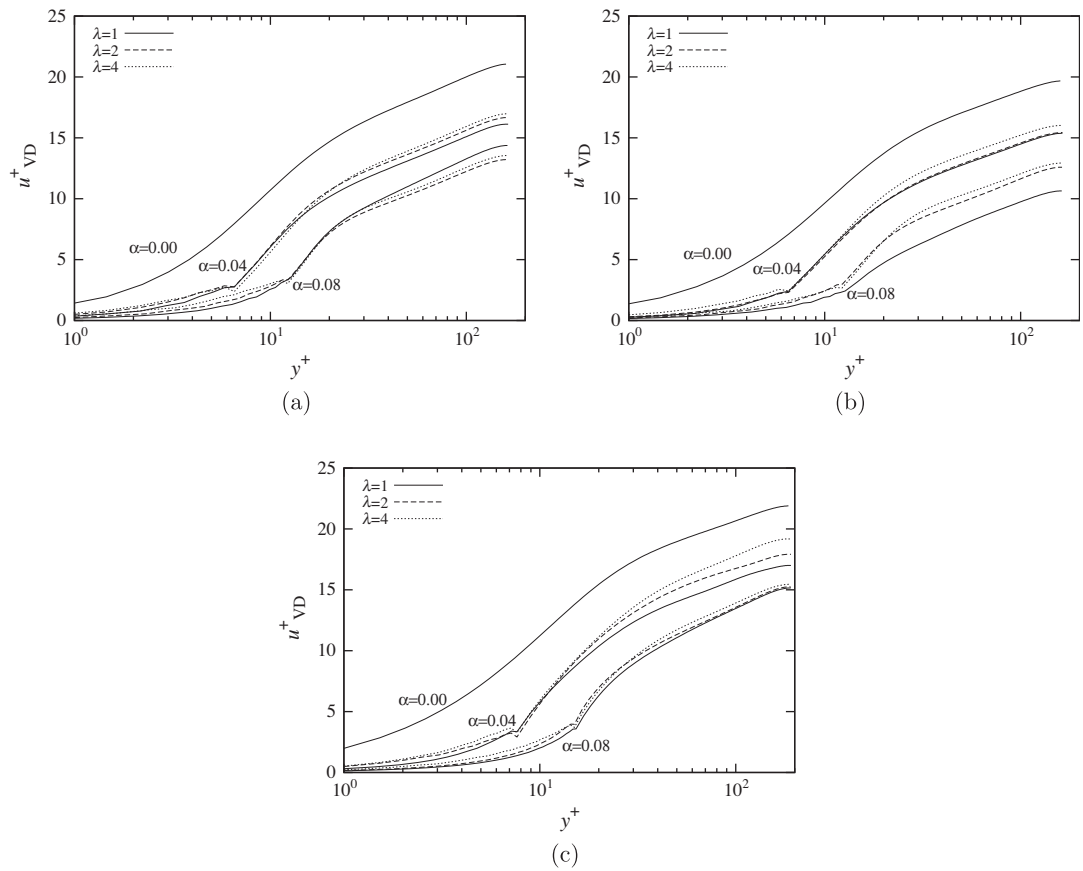


Fig. 16. Van Driest transformed velocity profiles for $M = 0.3$, $M = 1.5$, and $M = 3.0$.

sheltering effect to the following elements. The parameter used to distinguish between these two regimes is the solidity parameter, Λ, which is defined as the ratio of the total projected frontal roughness area to unit wall-parallel projected area. For the surfaces investigated here, this can be given by

$$\Lambda = \frac{2\alpha}{\lambda}. \tag{9}$$

If $\Lambda \gtrsim 0.15$ then the surface is considered to be of d-type, so that the roughness effect decreases with increasing solidity; whereas if $\Lambda \lesssim 0.15$ the surface is k-type and the roughness effect increases with the solidity. So, for the case S-1-08, with $\alpha = 0.08$ and $\lambda = 1$, the solidity is $\Lambda = 0.16$, therefore it is on the margin of the two regimes, whereas the $\lambda = 2$ and 4 cases are clearly in the k-type category. Therefore, depending how the flow interacts with the surface, it may behave as either k-type or d-type roughness. For the $M = 0.3$ case it appears that there is a significant mutual sheltering effect taking place, causing the roughness effect to decrease as λ is decreased. In contrast, the same case at $M = 1.5$ has a significant increase in roughness effect.

Owing to the multiple combinations of wavelengths and surface amplitudes that have been simulated it is useful to consider a single parameter that may describe the variations. The effective slope parameter (ES) as first used in Napoli et al. (2008) is to be used instead of the solidity parameter in this study. The reasoning for this is that, although the ES parameter yields similar results to the solidity, it takes into account the slope of the wall, whereas the solidity parameter is more commonly applied to square block roughness rather than wavy surfaces. Napoli et al. defined the ES as

$$\mathrm{ES} = \frac{1}{L_x}\int_{L_x}\left|\frac{\partial k}{\partial x}\right|\mathrm{d}x \tag{10}$$

where k is the surface height. They used this parameter to characterise randomly rough surfaces which vary only in the streamwise direction. With this they clearly demonstrated a dependence of velocity deficit on the effective slope parameter. Fig. 17 shows velocity deficit as a function of the ES parameter for the data gathered here. There is a clear division in the data that is dependent upon the surface amplitude. Looking for the impact of the Mach number, it is seen that in general the roughness effect is reduced with increased Mach number, as seen previously. There are some points that do not quite follow this trend, which will be discussed in the following paragraphs.

Fig. 18 shows the mean streamwise velocity profiles through the surface peaks and troughs for $\alpha = 0.04$ and $\alpha = 0.08$. The

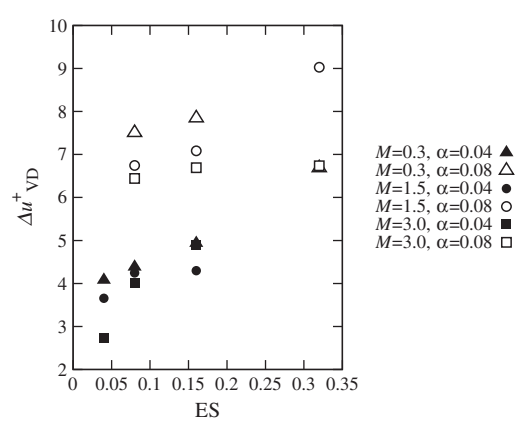
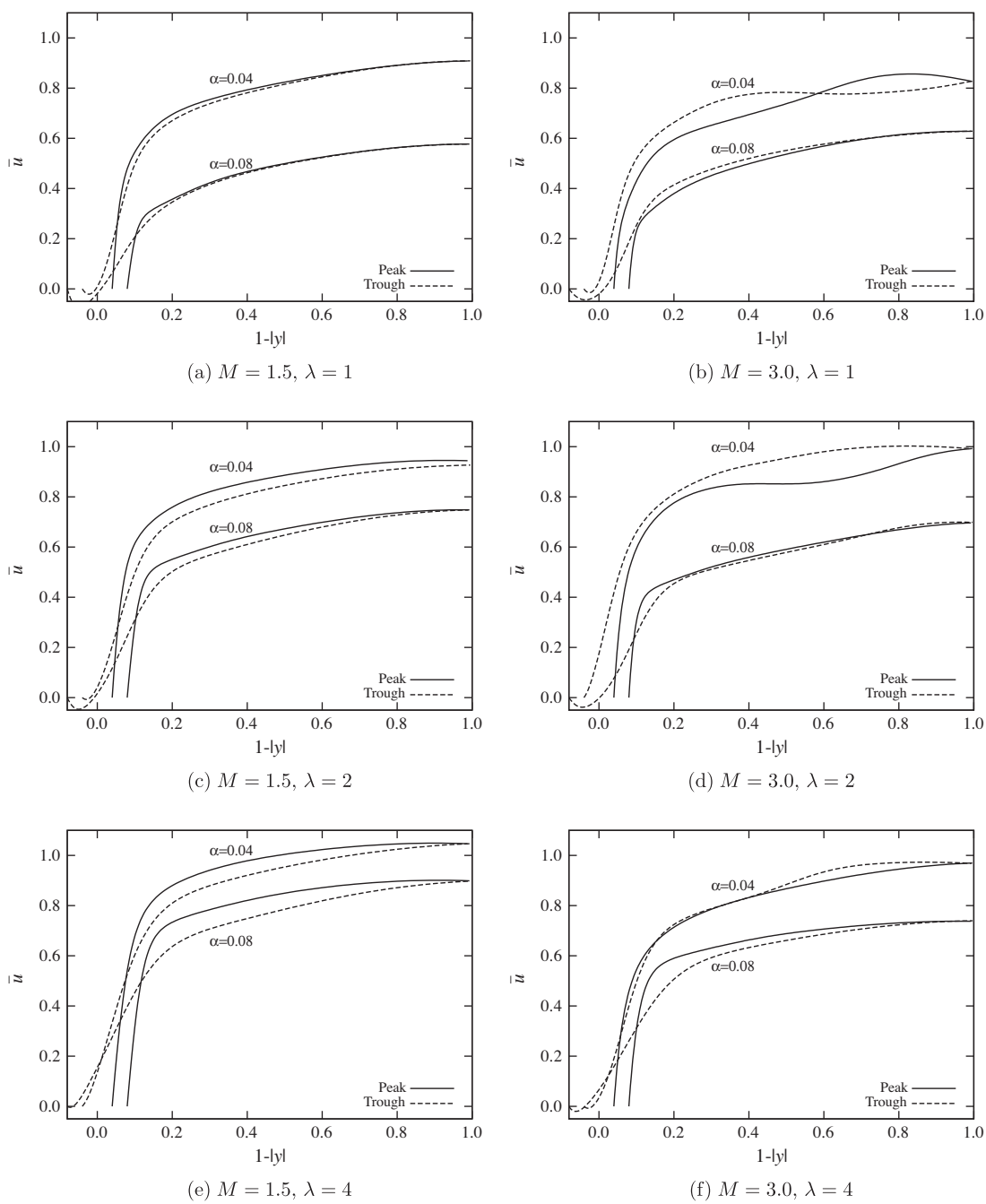


Fig. 17. The velocity deficit against the effective slope parameter.

$M = 0.3$ profiles have not been included since the profiles are very similar to the $M = 1.5$ cases. As with the previous amplitude study, it is clear that the velocity profiles for the $M = 3.0$ cases are significantly different. For cases H-1-04 and H-2-04 in Fig. 18b and d the reversal of flow behaviour is seen again, with the fluid over the trough moving faster than fluid over the peaks, in contrast to the $M = 1.5$ behaviour. Case H-1-04 indicates a velocity profile that has more local variations than the other cases due to peaks in the profiles occurring at different points. Case H-4-04 in Fig. 18f shows less variation compared to the shorter surface wavelengths, although towards the channel centreline some variation is seen in the flow over the trough. The profiles for the $\alpha = 0.08$ surfaces show much less variation with Mach number. Case H-1-08 shows the most change, with the fluid over the trough once again moving faster than over the peak, although this difference only occurs near the wall and the two profiles convergence quickly to the same value.

Fig. 19 shows the Favre-averaged fluctuations. As the Mach number is increased so too does the variation in the streamwise fluctuations, u'', across the entire channel. For the $M = 0.3$ cases in Fig. 19a and b the differences are limited to the near-wall region and all results converge by $y^+ = 40$. This point of convergence moves outwards to $y^+ = 80$ for $M = 1.5$ in Fig. 19c and d, although the profiles do not align as closely as in the lower Mach number case, especially for the $\alpha = 0.08$ case shown in Fig. 19d. The peak values for all cases at $M = 1.5$ are lower in comparison to $M = 0.3$, although the relative behaviour of the different surfaces is the same, in that cases L-1-08 and L-2-08 tend to be lowest with both



(a) $M = 1.5,\ \lambda = 1$ (b) $M = 3.0,\ \lambda = 1$

(c) $M = 1.5,\ \lambda = 2$ (d) $M = 3.0,\ \lambda = 2$

(e) $M = 1.5,\ \lambda = 4$ (f) $M = 3.0,\ \lambda = 4$

Fig. 18. Mean velocity, $\bar{u}$, profile slices at the peak and trough of the wall.

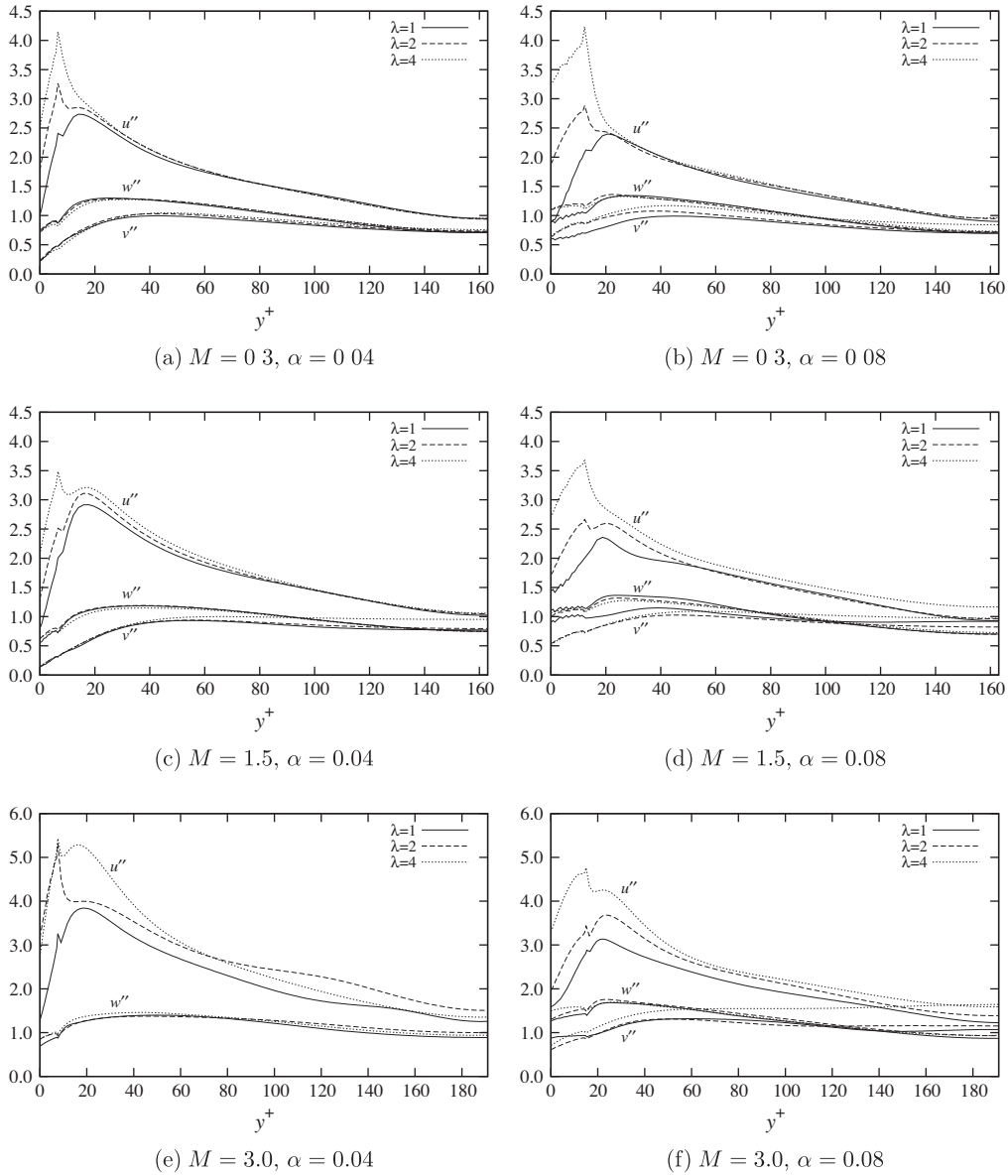


Fig. 19. Mean Favre-averaged velocity fluctuation profiles in wall units.

$\lambda = 4$ surfaces being the highest. For the $M = 3.0$ profiles in Fig. 19e and f there is greater variation in the values throughout the channel and the profiles do not converge in the outer layer, with certain cases such as H-4-04 and H-1-04 showing persistent wavy features.

The spanwise fluctuations, w'', in Fig. 19 show the same behaviour as the other fluctuating variables. As the Mach number is increased there is greater variation in the profiles throughout the channel. However, these variations are only small, although they occur in the near-wall region as well as at the channel centreline.

The wall-normal fluctuations, v'', for the $M = 0.3$ cases in Fig. 19a and b show the profiles to be very close, although case S-4-08 shows greater fluctuations across the channel. The $M = 1.5$ cases in Fig. 19c and d show more variation, with the $\alpha = 0.08$ surfaces having the largest near-wall fluctuations. The largest difference here is in case S-1-08, which is one of the lower profiles for $M = 0.3$, but at $M = 1.5$ has the largest peak value. The most striking differences is again the $M = 3.0$ profiles, shown separately in Fig. 20 for $\alpha = 0.04$. The change in surface wavelength has modified the nature of the large peaks, with the magnitude, location and number of peaks dependent on the number of surface waves. The $\alpha = 0.08$ surfaces in Fig. 19f do not show the same profile variations, although in case H-4-08 the peak value occurs at the channel centreline rather than near the wall as with the other cases. The fluctuation slices in Fig. 21 show that for the $M = 3.0$ cases the peaks are associated with shock waves in the channel.

The results seen so far suggest the presence of shock waves in the $M = 3.0$ case that alter the properties of the flow over a large area. The instantaneous divergence of velocity contours plotted in Figs. 22 and 23 shows that shock waves are indeed present in the channel. The shock angle is consistent across the different wavelengths as is the generation of shock waves per surface wavelength. However, as the number of surface waves is increased more shocks are generated per unit length increasing the number of shock interaction points, resulting in the differing number and locations of the peaks in the wall-normal velocity. It is noted that the strengths of the shock waves are also dependent upon the number of surface waves within the domain, since the shocks are strongest for the $\lambda = 4$ case and weakest when $\lambda = 1$. The recirculation region in H-4-04 is closely linked with the shock wave,

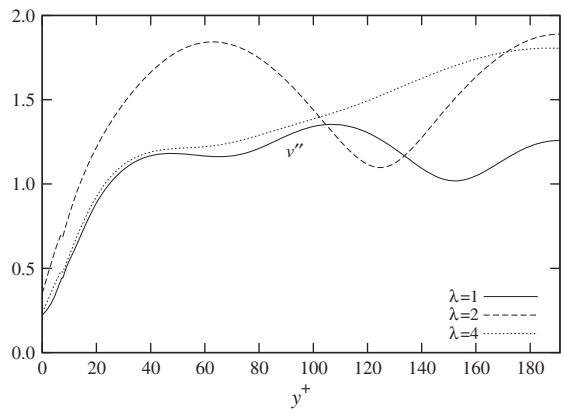


Fig. 20. Mean Fare-averaged wall-normal velocity fluctuation for $\alpha = 0.04$ at $M = 3.0$.

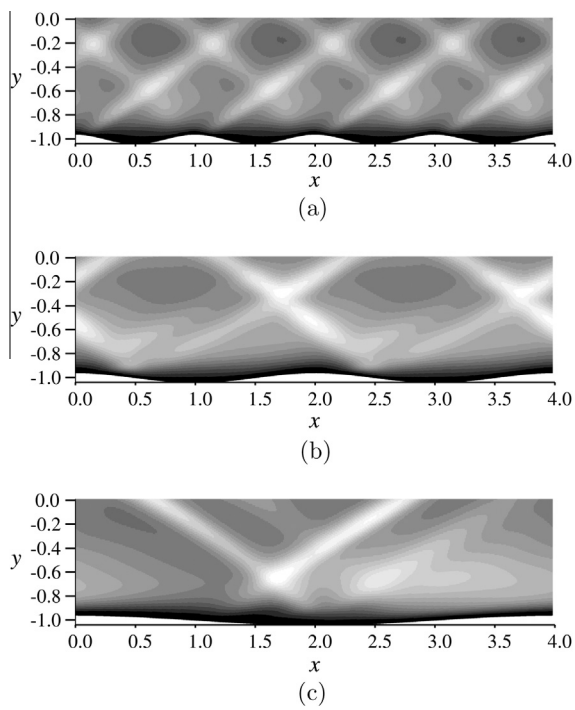


Fig. 21. Favre-averaged wall-normal velocity fluctuation slices in the xy-plane for (a) $\lambda = 1$, (b) $\lambda = 2$, and (c) $\lambda = 4$.

probably being created by a shock wave generated on the opposite wall, whilst also creating a reflected shock wave. As with the amplitude study, it is found that the increase in surface amplitude causes the formation of stronger, but more intermittent, shocks. Also the strength of the shocks is linked with the number of surface waves, as already noted. Figs. 24 and 25 show the mean $\nabla \cdot u$ for all the $M = 3.0$ cases. As with the amplitude study, the $\alpha = 0.04$ surface has stronger mean compressions then the $\alpha = 0.08$ surfaces. When changing the wavelength for the $\alpha = 0.04$ surfaces in Fig. 24 the shocks are still strongest for the $\lambda = 4$ case and are weakest for $\lambda = 1$. For the $\alpha = 0.08$ surfaces in Fig. 25 it is clear that the $\lambda = 2$ case has the strongest shocks, with only weak features observed in the $\lambda = 4$ and $\lambda = 1$ shows no clear shock formation.

Net pressure profiles are plotted in Figs. 26 and 27. As with the amplitude study, an increase in Mach number moves the location of the maximum pressure towards the surface peak. However, as noted previously, case H-2-04 does not conform to this trend.

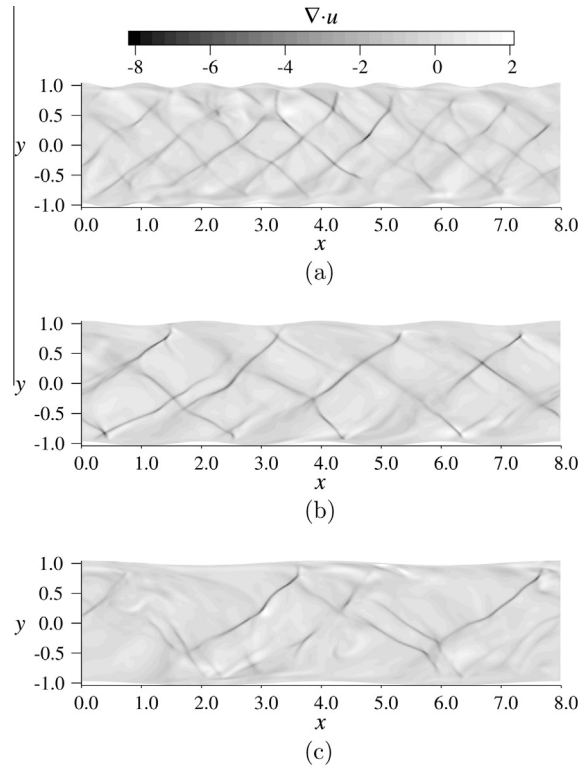


Fig. 22. Instantaneous $\nabla \cdot u$ contours for $\alpha = 0.04$ with (a) $\lambda = 1$, (b) $\lambda = 2$, and (c) $\lambda = 4$.

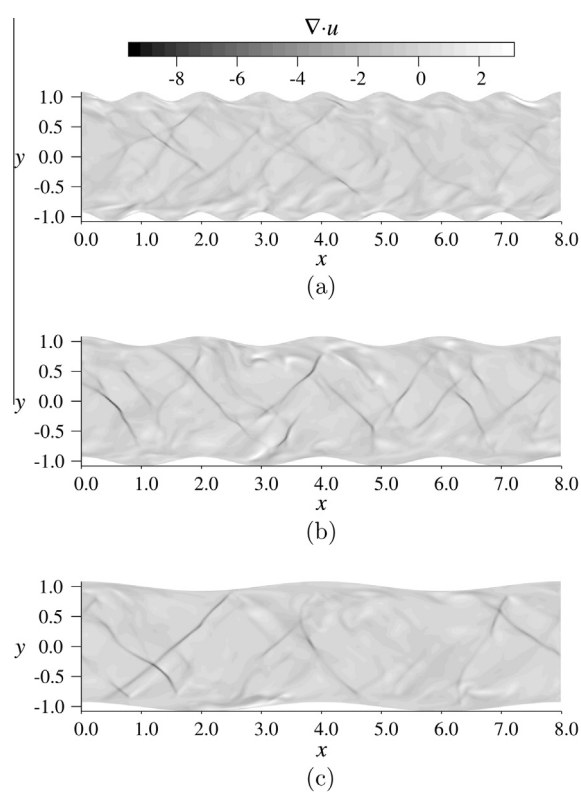


Fig. 23. Instantaneous $\nabla \cdot u$ contours for $\alpha = 0.08$ with (a) $\lambda = 1$, (b) $\lambda = 2$, and (c) $\lambda = 4$.

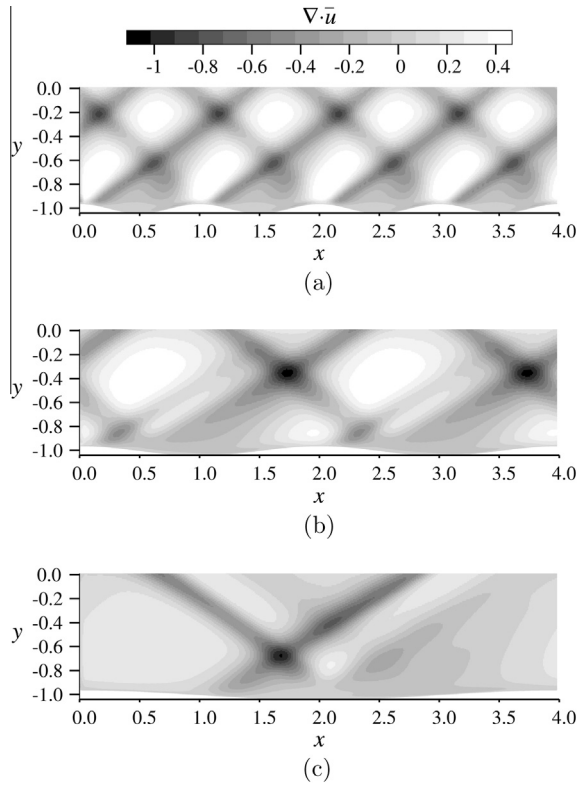


Fig. 24. Mean $\nabla \cdot u$ contours for $\alpha = 0.04$ with (a) $\lambda = 1$, (b) $\lambda = 2$, and (c) $\lambda = 4$.

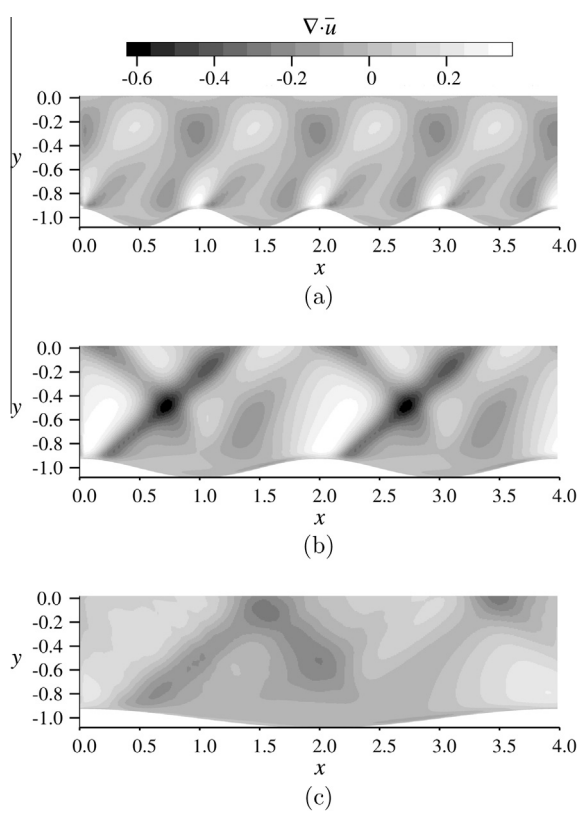


Fig. 25. Mean $\nabla \cdot u$ contours for $\alpha = 0.08$ with (a) $\lambda = 1$, (b) $\lambda = 2$, and (c) $\lambda = 4$.

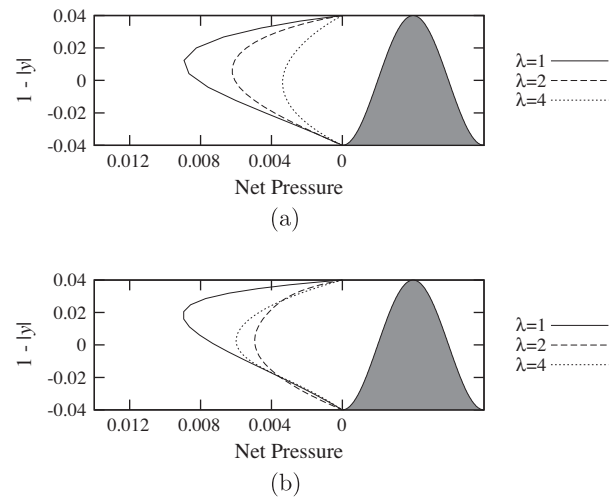


Fig. 26. Net pressure profiles for $\alpha = 0.04$ at (a) $M = 1.5$ and (b) $M = 3.0$.

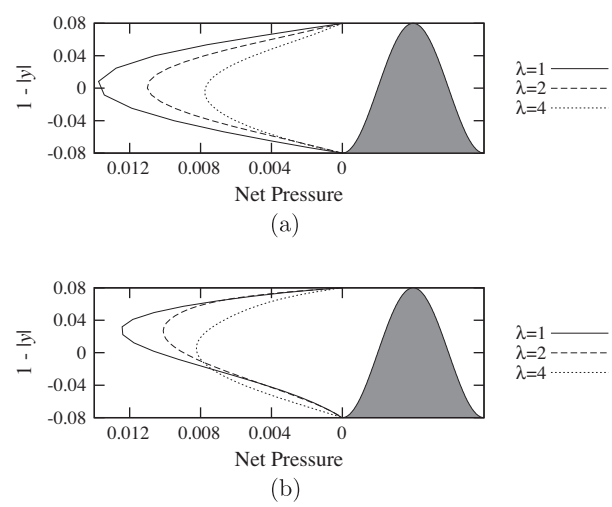


Fig. 27. Net pressure profiles for $\alpha = 0.08$ at (a) $M = 1.5$ and (b) $M = 3.0$.

The wavelength of the surface also affects the location of the peak pressure. As the surface wavelength is decreased, the peak pressure value increases and moves upwards, although this upward movement is not as large as the movement generated by the change in Mach number.

Fig. 28 shows the skin friction profiles for the $\alpha = 0.04$ surfaces. The $M = 1.5$ cases show a clear trend that, as the wavelength is reduced, the flow in the trough goes from being completely attached, as in case L-4-04, to undergoing a separation, which increases in relative size as the wavelength continues to reduce. Meanwhile, the peak skin friction increases as the wavelength is reduced, with the peak location again dependent on the wavelength. These trends are not clearly observed in the $M = 3.0$ cases. Here, the H-4-04 case undergoes separation, while the H-2-04 case does not. For the H-4-04 case there is a steep decrease in the skin friction leading up to the separation, which suggests that this separation is caused by the strong adverse pressure gradient imposed by the shock wave seen previously. Case H-1-04 is similar to the lower Mach number case, although the profile over the surface peak is flatter.

Skin friction profiles for the $\alpha = 0.08$ surfaces are plotted in Fig. 29. The $M = 1.5$ cases again show clear trends based upon the

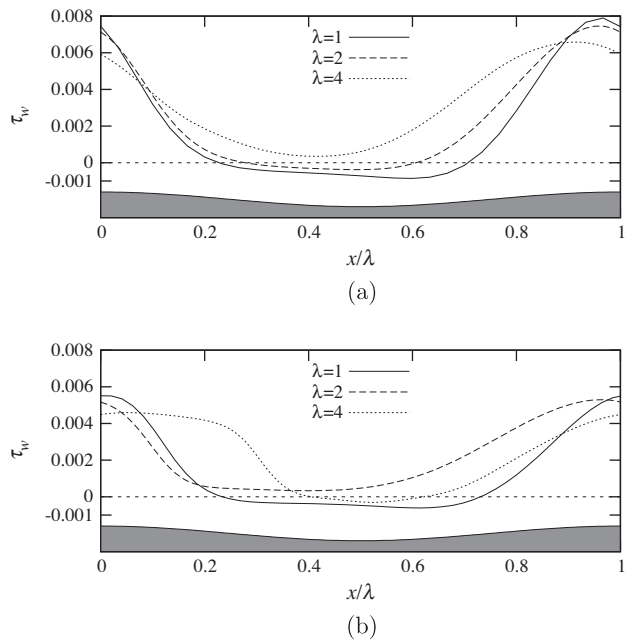


Fig. 28. Skin friction profiles for $\alpha = 0.04$ at (a) $M = 1.5$ and (b) $M = 3.0$.

surface wavelength. All three cases in Fig. 29a shows separation at approximately the same location, but the size of the recirculating region increases as the surface wavelength decreases. The peak skin friction does not have a clear trend with the surface wavelength, unlike at the lower Mach number.

Fig. 30 compares the percentage of the total drag on the surface attributed to the skin friction and pressure for all the cases with variable wavelength. There is a general pattern that as the Mach number or surface amplitude is increased, the flow becomes more dominated by the pressure force on the surface. Also, the pressure force becomes more dominant as the surface wavelength is decreased, but the difference in values here is reduced by increasing the Mach number. For $M = 0.3$ the pressure force varies by almost 55% when the wavelength is changed, as opposed to the 35%

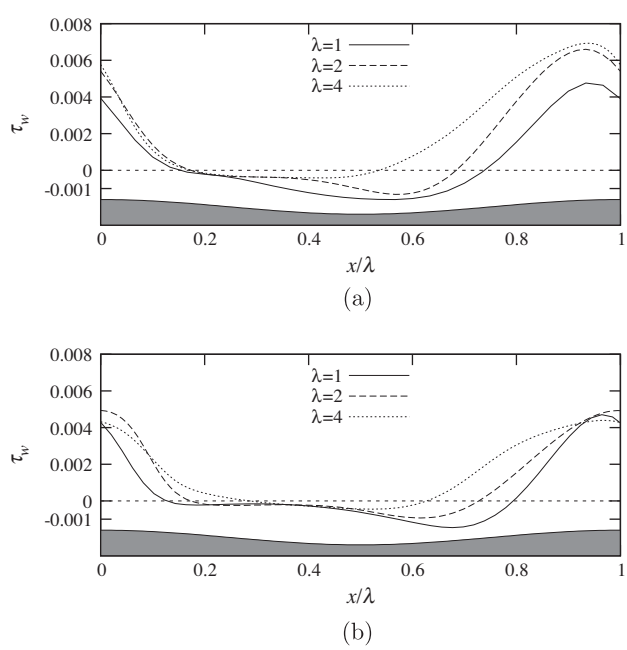


Fig. 29. Skin friction profiles for $\alpha = 0.08$ at (a) $M = 1.5$ and (b) $M = 3.0$.

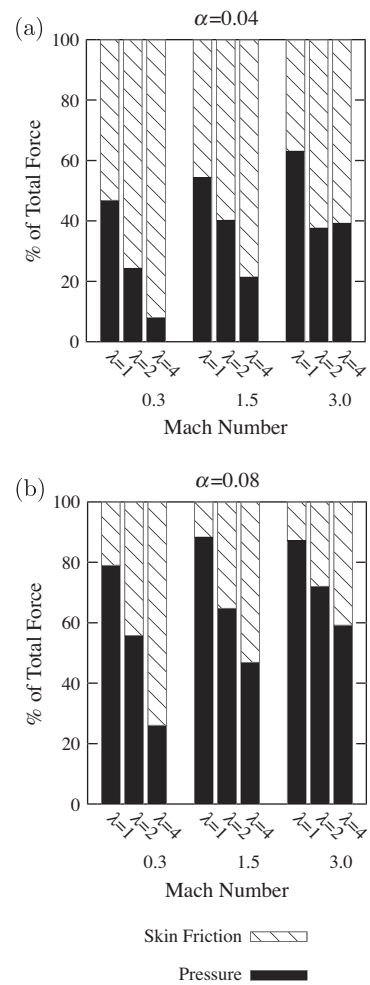


Fig. 30. Percentage of total drag force attributed to pressure and skin friction for (a) $\alpha = 0.04$ and (b) $\alpha = 0.08$.

variation seen at $M = 3.0$ for $\alpha = 0.08$. From this plot two anomalous cases can be seen. First, case H-2-04 has an unusually low pressure drag component, assumed to be linked to the lack of separation causing the skin friction to be relatively higher. Additionally, case H-1-08 shows a slight reduction in the pressure drag component compared to case L-1-08, but there is no clear explanation for this.

In summary, the surface wavelength effects are more complicated than the change in surface amplitude. It is important to consider the roughness regime when the wavelength is changed, since the elements may begin to shelter each other in certain circumstances. In general, the change in wavelength is a secondary effect on the flow in comparison with the surface amplitude, but the surface wavelength does determine the number and location of shock wave interaction points. The effects of the change in surface wavelength are diminished by increasing the Mach number, as with the effect of surface amplitude. Although properties such as the velocity deficit see a reduced effect, it is important to consider how other properties, such as the velocity fluctuations, change across the entire channel at high Mach numbers.

4. Conclusions

Results from a series of DNS have been presented that have considered the effect on turbulent channel flow of 2D surface wave amplitude and wavelength in combination with Mach number. The cases simulated cover transitionally and fully rough surfaces.

The results show that for moderate surface amplitudes there are strong compressibility effects. In general the effect of the surface appears to be reduced at high Mach number when considering the velocity deficit of a van Driest transformed velocity profile. However, considering just this parameter conceals many features in the flow. Shock waves were found within the domain for all the Mach 3.0 cases, although the strength and number of these was heavily dependent upon the surface amplitude and the number of surface peaks. At the lower surface amplitude, the shocks were found to be more persistent than for the higher amplitude, although they were also weaker. The effect of the surface wavelength appeared to be mainly to modify the number and location of shock wave interaction points, with a small influence on the shock strength.

As a result of these regular shock waves, mean flow and turbulence properties were found to be modified. The mean streamwise velocity behaviour was found to reversed, with the higher velocity fluid occurring over troughs rather than the peaks of the surface. The most dramatic change occurred in the Favre-averaged wall-normal velocity fluctuations, where large peaks were formed throughout the channel, which were located around the mean shock location, particularly near shock interaction points.

The forces acting upon the surfaces were also modified. When the Mach number or surface amplitude was increased, so too was the contribution of the pressure force towards the total drag acting upon the surface, while reducing the surface wavelength also had the same effect. The location where the peak pressure acts upon the surface moved upwards towards the surface peak for an increase in Mach number as well as for a decrease in wavelength. Shock-induced recirculation regions were also found, which contributed to a significant reduction in the skin friction drag for the surfaces where this occurred.

Acknowledgements

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